Adaptive Wind Power Forecasting Using Concept Drift Detection

Viva Voce & Technical Presentation Guide — Simplified Reference

Institution: Panimalar Engineering College (Anna Univ.)

Degree: B.Tech Artificial Intelligence & Machine Learning

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Target Horizon: 1-Hour-Ahead Generation on 2.0 MW Wind Farm

1. The Big Picture & Fundamental Physical Challenge

Power grid operators face a critical challenge: electricity grids must maintain instantaneous balance between generation and demand. Unlike thermal generators whose fuel intake is throttled at will, wind turbines generate power only when the wind blows. Grid dispatchers require an accurate **1-hour-ahead forecast** to schedule operating reserves and avoid severe grid imbalance fines.

The Cubic Law Amplifier: Wind kinetic power follows $P = ½ \rho A Cp v^3$. Because electrical yield scales with the cube of wind speed, a modest **10% error in forecast wind speed amplifies into a ~33% discrepancy in forecasted power**. Consequently, even small model inaccuracies produce large power errors.

2. What is Concept Drift & Why Do Standard Models Fail?

Traditional machine learning assumes *stationarity*—that relationships learned during training remain unchanged forever. In real wind farms, this assumption fails due to:

- **Turbine Blade Derating & Ageing:** Surface roughness, insect accretion, and gearbox wear mean the same wind yields less power over time.
- **Directional Wake Losses:** Construction of new turbines or buildings creates directional aerodynamic turbulence (e.g. 200°–265° wake sector).
- **Sensor Recalibration (Virtual Drift):** Anemometer replacement shifts recorded speeds without altering actual physical wind energy.

The Danger of Static Models: A model trained once becomes silently outdated. It continues issuing forecasts based on invalid rules, but has zero awareness that its accuracy has deteriorated.

3. The Trap of Scheduled Calendar Retraining (e.g. Weekly)

The standard industry practice is blind calendar retraining (e.g., retrain every Sunday). Our research demonstrates that this is flawed:

- **Sluggish Reaction:** An abrupt wake effect occurring on Monday leaves the grid with 6 full days of degraded forecasts before the next retrain.
- **Harmful When Stationary:** In stationary weather, restricting training to a short 4-week window causes high variance, making errors **30.2% worse** than leaving the static model alone.
- **Wasted Computation:** Retraining 80 times on a calendar costs 4.9× more compute with no empirical benefit.

4. The Proposed Solution (4-Step Pipeline)

Step	Process & Method	Technical Innovation & Safeguard
1. Predict	1-Hour-Ahead Forecast via HistGradientBoostingRegressor using 10 features (v, sin/cos direction, temp, power lags).	Strict chronological time-ordering: guarantees **zero future target leakage**.
2. Standardize	Convert raw residual into scale-free score: $z(t) = \|y(t) - ■(t)\| / m(■(t))$ using 6 forecast-level quantile bins.	Removes wind-level error heteroscedasticity; prevents false alarms during normal winter storms.
3. Detect	Sequential Page-Hinkley test on z(t) with threshold $\lambda = 15\sigma■ = 12.0$ and tolerance $\delta = 0.6\sigma■ = 0.48$.	Pinpoints exact change point $\tau■ = argmin(m_i)$; input KS/PSI tests act strictly as advisory channels.

4. Adapt	Adaptive Retraining Controller fits on recent 4-week (672 h) window with recency decay (half-life 336 h).	Downweights pre-change observations (0.08×) and enforces **48-hour refractory cooldown**.

5. Empirical Benchmark Results (10 Random Seeds, 13,440 Hourly Steps Each)

Metric	Static Model	Adaptive Framework	Weekly Retrain	Viva Defence Evidence
Drift MAE (kW)	134.7 ± 5.6	**33.7 ± 2.1**	35.3 ± 1.3	**75.0% error reduction** (paired t = 63.5, p = 3.0×10■[13])
Drift RMSE (kW)	178.4 ± 5.6	**53.0 ± 3.7**	62.7 ± 5.4	**70.3% error reduction** (paired t = 5.8, p = 2.2×10■■)
nRMSE (%)	8.92 ± 0.28%	**2.65 ± 0.19%**	3.14 ± 0.27%	Reduced by 6.27 percentage points of 2.0 MW capacity
Retrainings	0.0	**16.2 ± 6.6**	80.0	**4.9× fewer retrainings** than scheduled weekly updates
Compute Time (s)	0.00 s	**1.98 ± 0.82 s**	9.59 ± 0.53 s	**4.9× less CPU compute**; monitoring costs ~0.049 ms/h
No-Drift MAE	25.5 ± 0.6 kW	**25.5 ± 0.6 kW**	33.2 ± 1.5 kW	Adaptive raises 0 alarms; weekly suffers **+30.2% penalty**
Detection Latency	—	**21.5 h mean**	Up to 168 h	10/10 detected; **zero false alarms** before onset

6. Tour of the 7 Web Dashboard Pages (Live Demonstration)

- **Page 1 — Overview Dashboard:** Real-time master control room with live generation gauges, latest 1h forecast, active MAE/RMSE, drift status badge (Stable/Warning/Drift Detected), and actual vs predicted power curve.
- **Page 2 — Forecasting Studio:** Interactive workbench to adjust sliders (wind speed, direction, temperature, power lags) and compute immediate 1h predictions with 90% confidence intervals and 24h residual breakdown.
- **Page 3 — Drift Monitoring:** Visualizes stream playback with **Step 1h**, **Step 24h**, and **Reset** controls. Charts Page-Hinkley cumulative statistic against threshold $\lambda = 12.0$ alongside daily KS and PSI monitors.
- **Page 4 — Adaptive Learning:** Details model adaptation: displays active model version (v1, v2..), 48h cooldown countdown timer, sample weights curve (336h half-life + 0.08x pre-change weight), and manual retrain modal.
- **Page 5 — Model Comparison:** Head-to-head bar charts of Static vs Adaptive vs Weekly models, paired Student's t-test and Wilcoxon signed-rank tables ($p < 0.05$), and automated analytical interpretations.
- **Page 6 — Dataset Explorer:** Ingests synthetic or custom CSV SCADA data; features paginated table, missing-value health, wind speed vs power scatter plot with empirical curve, 16-sector wind rose, and Pearson correlation heatmap.
- **Page 7 — Experiments & Reports:** Research laboratory allowing examiners to test seeds 0–9, drift vs no-drift, sensitivity curves (threshold λ, cooldown, window size, magnitude), and export formatted PDF reports.

7. Viva Voce Examiner Q&A; Defense Sheet

Q1: Why did you choose HistGradientBoostingRegressor over deep learning (e.g. LSTM)?

Defense: A: HistGBR trains in ~0.12 seconds on tabular data using histogram binning. When retraining repeatedly in stream environments, deep learning requires heavy GPU compute, extensive tuning, and risks catastrophic forgetting. HistGBR provides top tabular accuracy with negligible retraining overhead.

Q2: Why not trigger retraining whenever the Kolmogorov-Smirnov (KS) test detects wind speed changes?

Defense: A: Wind speed is strongly autocorrelated (AR(1) phi = 0.96). Ordinary seasonal shifts yield KS p-values of 10■[1]■ even without any turbine changes. Retraining on input shifts alone caused >150 false alarms per run. Input tests must act strictly as advisory channels corroborated by elevated prediction error.

Q3: What prevents the system from retraining constantly when a drift occurs?

Defense: A: Two safeguards: (1) A 48-hour refractory cooldown period prevents thrashing, and (2) The Page-Hinkley detector resets immediately upon alarm, requiring fresh accumulated evidence before another update is permitted.

Q4: Why does weekly retraining perform 30.2% worse than the static model in stationary weather?

Defense: A: Restricting training to a short 4-week window starves the model of rare high-wind states captured in the larger historical window, increasing estimation variance. Scheduled retraining without evidence is actively harmful.

30-Second Viva Elevator Pitch:
"Our project develops an intelligent 1-hour-ahead wind power forecasting system that detects when turbine degradation or directional wake effects cause predictions to deteriorate. Instead of retraining blindly on a calendar schedule, our system uses scale-standardized residual monitoring and the Page-Hinkley sequential test to retrain only when statistical evidence proves the model is outdated. This achieves a ***75% error reduction*** *under concept drift while requiring* ***4.9× less computation*** *than traditional scheduled retraining."*